

Network Security

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✅ PART 1 — TOP 5 IMPORTANT 5-MARK QUESTIONS

◆ Q1. (Chapter-1 Introduction to Network Security)

Explain CIA Triad with suitable diagram and examples.

◆ STEP-1: THEORY EXPLANATION (FROM ZERO LEVEL)

★ Intuition (Simple Idea)

When we protect any data, we mainly care about **3 things**:

1. Nobody reads it without permission
2. Nobody changes it
3. It should always be available

This is called **CIA Triad**.

★ Basic Explanation

CIA stands for:

- **C → Confidentiality**
- **I → Integrity**
- **A → Availability**

These are the **main goals of network security**.

★ Exam-Level Definition

The CIA triad is a fundamental model used to guide policies for information security consisting of confidentiality, integrity, and availability.

★ Real-Life Analogy

Think of **bank locker**:

- Confidentiality → Only owner opens it
 - Integrity → Nobody changes gold inside
 - Availability → You can access locker anytime
-

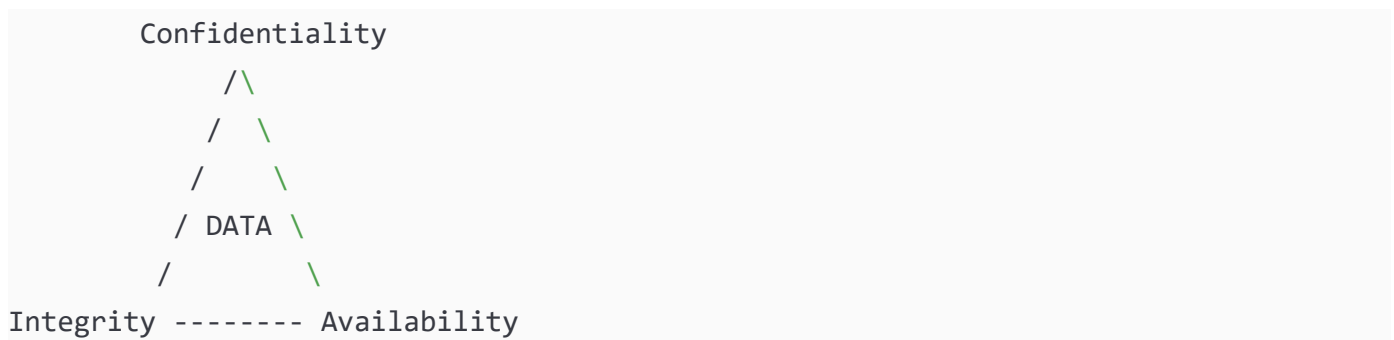
★ Practical Use in Network Security

- Online banking → needs confidentiality
 - Software update → needs integrity
 - Google servers → need availability
-

★ Important Exam Writing Points

- CIA is foundation of all security systems
 - Every attack targets one or more CIA elements
 - Used in OSI security architecture
-

◆ STEP-2: DIAGRAM (MANDATORY)



🔍 Label Explanation

- Triangle center = Protected data
 - Confidentiality side = secrecy
 - Integrity side = correctness
 - Availability side = accessibility
-

◆ STEP-3: FORMULAS (SYMBOL MEANING)

No heavy math, but concept symbols:

Confidentiality → Encryption → $C = E(K, P)$

Where:

E = Encryption

K = Key

P = Plaintext

C = Ciphertext

◆ STEP-4: NUMERICAL

Not applicable.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

CIA triad consists of Confidentiality, Integrity, and Availability. Confidentiality ensures data secrecy, integrity ensures data is not modified, and availability ensures data is accessible when needed. It forms the foundation of network security.

◆ STEP-6: EXAM TIP

- Draw triangle diagram (very important)
 - Write 1 example for each CIA
 - Use keywords: secrecy, correctness, access
 - Don't forget real-life analogy
-

◆ Q2. (Chapter-2 TCP/IP Protocol Suite)

Explain TCP/IP layered architecture with diagram.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Internet works like **post office system**.
Different layers handle different tasks.

★ Basic Explanation

TCP/IP has **5 layers**:

1. Application
 2. Transport
 3. Network
 4. Data Link
 5. Physical
-

★ Exam Definition

TCP/IP is a layered networking model that defines how data is transmitted over the Internet using hierarchical protocols.

★ Real-Life Analogy

Sending parcel:

- Write message → Application
 - Packaging → Transport
 - Address routing → Network
 - Road delivery → Data link
 - Wires → Physical
-

★ Practical Use

Used in:

- Internet communication
 - Web browsing
 - Email systems
-

★ Important Exam Points

- TCP = reliable
 - UDP = fast but unreliable
 - IP = addressing
-

◆ STEP-2: DIAGRAM

```
+-----+
|  Application Layer  | → HTTP, FTP, SMTP
+-----+
|  Transport Layer   | → TCP, UDP
+-----+
|  Network Layer     | → IP, ICMP
+-----+
|  Data Link Layer   | → Ethernet
+-----+
|  Physical Layer    | → Cables, signals
+-----+
```

🔍 Label Explanation

- Application → User services

- Transport → End-to-end delivery
 - Network → Routing packets
 - Data link → Frame transfer
 - Physical → Bit transmission
-

◆ STEP-3: FORMULAS

No math.

◆ STEP-4: NUMERICAL

Not required.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

TCP/IP model consists of 5 layers: Application, Transport, Network, Data Link, and Physical. Each layer performs a specific function for data communication over the Internet.

◆ STEP-6: EXAM TIP

- Always draw stacked layer diagram
 - Mention example protocols
 - Write TCP vs UDP difference
 - Very common PYQ
-

◆ Q3. (Chapter-3 General Categories of Security Attacks)

Differentiate between passive and active attacks with examples.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Some attackers only **listen**, others **damage**.

★ Basic Explanation

Two main attack types:

- Passive → Monitoring

- Active → Altering
-

★ Exam Definition

Passive attacks monitor information without altering data, while active attacks modify, fabricate, or disrupt data transmission.

★ Real-Life Analogy

Passive → CCTV spying

Active → Breaking door

★ Practical Use

Passive:

- Traffic analysis

Active:

- DoS attack
 - Data modification
-

★ Exam Points

Passive = hard to detect

Active = easier to detect

◆ STEP-2: DIAGRAM

Passive Attack:

Sender -----> Attacker (listens) -----> Receiver

Active Attack:

Sender --> Attacker (modifies data) --> Receiver

🔍 Label Explanation

- Passive attacker → only observes
 - Active attacker → modifies data
-

◆ STEP-3: FORMULAS

None.

◆ STEP-4: NUMERICAL

Not applicable.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Passive attacks involve monitoring data without modification (e.g., traffic analysis). Active attacks involve modification, fabrication, or disruption of data (e.g., DoS, replay attacks).

◆ STEP-6: EXAM TIP

- Write 2 examples each
 - Draw simple flow diagram
 - Mention detection difficulty
 - PYQ favorite
-

◆ Q4. (Chapter-4 Digital Signature)

Explain digital signature process with diagram.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Digital signature = **electronic handwritten signature**.

It proves:

- Who sent message
 - Message not changed
-

★ Basic Explanation

Uses:

- Hash function
 - Public-key cryptography
-

★ Exam Definition

A digital signature is a cryptographic technique that provides authentication, integrity, and non-repudiation using a private-public key pair.

★ Real-Life Analogy

Courier signed document.

★ Practical Use

- Email signing
- Software distribution
- Blockchain

★ Exam Points

- Private key signs
 - Public key verifies
-

◆ STEP-2: DIAGRAM

Sender (Alice)

Message M

|

v

Hash(M)

|

v

Sign with Private Key

|

v

Signature S

|

Send (M + S)

Receiver (Bob)

Hash(M)

|

Verify using Public Key

|

Valid / Invalid

🔍 Label Explanation

- Hash(M) → digest
 - Private key → signing
 - Public key → verification
-

◆ STEP-3: FORMULAS

Signature:

$S = \text{Encrypt_private}(H(M))$

Verification:

$H(M) = \text{Decrypt_public}(S)$

Where:

H = Hash function

M = Message

S = Signature

◆ STEP-4: NUMERICAL

Conceptual only.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Digital signature uses a hash function and private key to sign a message. The receiver verifies it using the sender's public key to ensure authenticity and integrity.

◆ STEP-6: EXAM TIP

- Draw full signing + verification flow
- Write 3 services: integrity, authentication, non-repudiation
- Common 5-mark question

◆ Q5. (Chapter-5 Cryptographic Hash Functions)

Explain properties of cryptographic hash functions.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Hash = Digital fingerprint.

★ Basic Explanation

Hash converts any size data → fixed-size digest.

★ Exam Definition

A cryptographic hash function is a one-way function that maps arbitrary input to fixed-length output while satisfying security properties.

★ Real-Life Analogy

Fingerprint → unique identity.

★ Practical Use

- Password storage
 - Blockchain
 - Digital signatures
-

★ Important Properties

1 Preimage Resistance

Cannot find input from hash.

2 Second Preimage Resistance

Cannot find another message with same hash.

3 Collision Resistance

Cannot find two different inputs with same hash.

◆ STEP-2: DIAGRAM

Message M ----> Hash Function ----> Digest $H(M)$

$M1 \neq M2$ but $H(M1) = H(M2) \rightarrow$ Collision

◆ STEP-3: FORMULAS

$H(M) \rightarrow$ Message digest

Where:

H = Hash function

M = Message

◆ STEP-4: NUMERICAL

Example:

Input: "HELLO"

Output: SHA-256 → fixed 256-bit hash

HELLO → HELL0 → totally different hash.

👉 FINAL ANSWER:

Explain message integrity using hash function with diagram.

★ Intuition

Receiver must know:

👉 Was message changed or not?

This is **message integrity**.

We use **hash functions** to protect integrity.

Sender sends:

- Message M

- Hash $H(M)$

Receiver recomputes hash and compares.

★ Exam Definition

Message integrity ensures that data has not been altered during transmission and is typically verified using cryptographic hash functions.

★ Real-Life Analogy

Parcel seal:

- If seal broken → tampered

★ Practical Use

- Software downloads
- Email verification
- Blockchain

★ Important Exam Points

- Hash = fingerprint
- Even 1-bit change → new hash

◆ STEP-2: DIAGRAM

Sender:

Message M

|

v

Hash Function

|

$H(M)$

|

Send (M + $H(M)$)

Receiver:

Receive M

|

Hash(M)

|

Compare with received $H(M)$

|

Match → Integrity OK

No match → Tampered

Label Explanation

- $H(M)$ = message digest
- Compare step = integrity verification

◆ STEP-3: FORMULAS

$H(M)$ → Message Digest

Integrity check:

$H_{\text{received}} = H_{\text{calculated}}$

If equal → valid

◆ STEP-4: NUMERICAL EXAMPLE

Message: HELLO

Hash: $H(\text{HELLO}) = A1B2$

If attacker changes:

HELLO → HELLo

New hash:

$H(\text{HELLo}) = 9F33$

Mismatch → tampering detected.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Message integrity ensures that transmitted data remains unchanged and is verified using hash functions by comparing computed and received message digests.

◆ STEP-6: EXAM TIP

- Always draw sender-receiver diagram
- Write comparison step
- Use word “digest”
- Very common PYQ

◆ Q7. (Chapter-7 Entity Authentication)

Explain challenge-response authentication with diagram.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Server asks:

👉 “Prove you know secret without telling me.”

This is challenge-response.

★ Basic Explanation

Steps:

1. Verifier sends random challenge
2. User processes it using secret
3. Sends response

If correct → authenticated.

★ Exam Definition

Challenge-response authentication verifies identity by requiring a claimant to correctly respond to a time-varying challenge using a secret.

★ Real-Life Analogy

Security question:

- Guard asks question
 - Only real person answers correctly
-

★ Practical Use

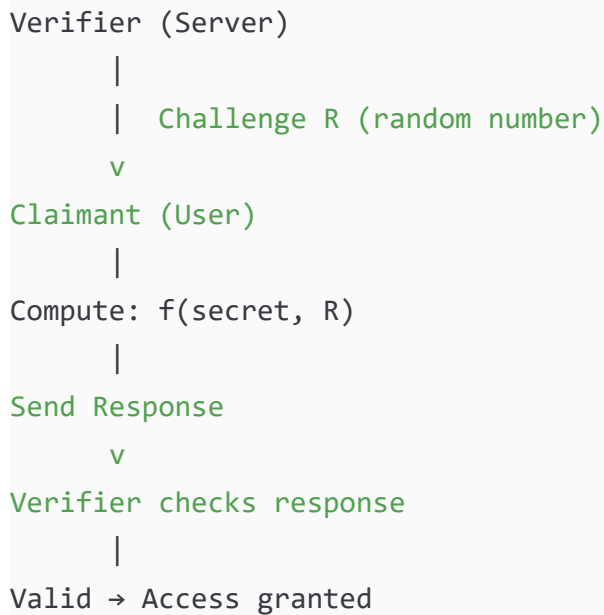
- ATM authentication
 - Two-factor login
 - Kerberos
-

★ Exam Points

- Uses nonce (random number)

- Prevents replay attacks

◆ STEP-2: DIAGRAM



🔍 Label Explanation

- R = nonce (random value)
- $f(\text{secret}, R)$ = response function

◆ STEP-3: FORMULAS

Response = $f(K, R)$

Where:

K = Secret key

R = Random challenge

◆ STEP-4: NUMERICAL EXAMPLE

Secret key K = 5

Challenge R = 7

Simple function:

Response = $K \times R = 35$

Server checks → authentic.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Challenge-response authentication verifies identity by sending a random challenge and validating the computed response using a shared secret.

◆ STEP-6: EXAM TIP

- Draw 3-step flow
 - Write nonce definition
 - Mention replay protection
 - PYQ favorite
-

◆ Q8. (Chapter-7 Entity Authentication)

Explain Lamport One-Time Password (OTP) mechanism with diagram.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Password should be used only once.

That is OTP.

★ Basic Explanation

Lamport OTP uses **hash chains**.

Steps:

1. Choose password P
2. Apply hash multiple times
3. Store last hash on server

Each login sends previous hash.

★ Exam Definition

Lamport OTP is a one-time password scheme that uses repeated hashing to generate a sequence of passwords valid only once.

★ Real-Life Analogy

Scratch cards:

- Use one code once

★ Practical Use

- Secure login
- Banking OTP systems

★ Exam Points

- Uses hash chains
- Prevents password reuse attacks

◆ STEP-2: DIAGRAM

User selects password P

Compute:

$h(P) \rightarrow h^2(P) \rightarrow h^3(P) \rightarrow h^4(P)$

Server stores $h^4(P)$

Login 1:

User sends $h^3(P)$

Server hashes $\rightarrow h^4(P)$ ✓

Login 2:

User sends $h^2(P)$

🔍 Label Explanation

- h = Hash function
- $h^4(P)$ = stored value

◆ STEP-3: FORMULAS

Hash chain:

$h^n(P) = h(h(\dots h(P)\dots))$

Where:

n = number of iterations

◆ STEP-4: NUMERICAL EXAMPLE

Let hash = +1 (simple demo)

$P = 10$

$h(P) = 11$

$h^2(P) = 12$

$h^3(P) = 13$

Server stores 13

User sends 12 → valid.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Lamport OTP uses repeated hashing to generate a chain of passwords, where each password is used once and verified using stored hash values.

◆ STEP-6: EXAM TIP

- Draw hash chain
 - Write “one-time use”
 - Mention replay protection
 - Important for authentication unit
-

◆ Q9. (Chapter-9 Denial of Service Attack)

Explain SYN flood attack with diagram.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Server opens many connections but attacker never completes them.

Server becomes busy → crash.

★ Basic Explanation

Uses TCP 3-way handshake.

Normal handshake:

1. SYN
2. SYN-ACK
3. ACK

Attacker sends only SYN repeatedly.

★ Exam Definition

A SYN flood is a DoS attack that exploits the TCP handshake by sending multiple SYN requests without completing the connection.

★ Real-Life Analogy

Calling customer care and hanging up repeatedly.

★ Practical Use

- Website crashes
 - Server overload
-

★ Exam Points

- Half-open connections
 - Memory exhaustion
-

◆ STEP-2: DIAGRAM

Normal:

Client → SYN → Server

Server → SYN-ACK → Client

Client → ACK → Connected

SYN Flood:

Attacker → SYN → Server

Server → SYN-ACK → (no reply)

Many half-open connections

Server resources exhausted

🔍 Label Explanation

- SYN = connection request
 - Half-open = incomplete handshake
-

◆ STEP-3: FORMULAS

No formulas.

◆ STEP-4: NUMERICAL EXAMPLE

Server can hold 100 connections.

Attacker sends:

1000 SYN requests.

Result:

Queue full → legit users blocked.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

SYN flood attack overwhelms a server by sending multiple incomplete TCP handshake requests, exhausting resources and causing denial of service.

◆ STEP-6: EXAM TIP

- Draw handshake diagram
- Write “half-open connections”
- Mention TCP handshake
- Very common exam question

◆ Q10. (Chapter-5 Cryptographic Hash Functions)

Explain Merkle-Damgård construction with diagram.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Long message → process block by block.

Like assembly line.

★ Basic Explanation

Merkle-Damgård builds hash functions using compression functions.

Steps:

1. Split message into blocks
 2. Process sequentially
 3. Produce final hash
-

★ Exam Definition

Merkle-Damgård construction is an iterative method of building hash functions by applying a compression function repeatedly over message blocks.

★ Real-Life Analogy

Factory conveyor belt:

Each station processes part of product.

★ Practical Use

Used in:

- SHA family
 - MD5
-

★ Exam Points

- Iterative hashing
 - Initial vector (IV)
-

◆ STEP-2: DIAGRAM

Message → M1 | M2 | M3 | M4

Initial Value (IV)

|

v

[Compression f]

|

H1

|

[Compression f] + M2

|

H2

|

... continues ...

|

Final Hash

🔍 Label Explanation

- M1, M2 = message blocks
- IV = initial value

- f = compression function
-

◆ STEP-3: FORMULAS

$$H_i = f(H_{i-1}, M_i)$$

Where:

H_i = intermediate hash

M_i = message block

f = compression function

◆ STEP-4: NUMERICAL EXAMPLE

Let:

$$IV = 0$$

$$\text{Blocks} = 2, 3$$

Simple function:

$$f(a, b) = a + b$$

Step 1:

$$H_1 = 0 + 2 = 2$$

Step 2:

$$H_2 = 2 + 3 = 5$$

Final hash = 5

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Merkle-Damgård construction creates hash functions by iteratively applying a compression function over message blocks starting from an initial value.

◆ STEP-6: EXAM TIP

- Draw block pipeline diagram
 - Write formula $H_i = f(H_{i-1}, M_i)$
 - Mention SHA usage
 - Very important theory question
-
-
-
-
-
-

◆ Q11. (Chapter-4 Digital Signature)

Differentiate between MAC and Digital Signature with examples.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Both MAC and Digital Signature verify authenticity.

But difference = **key usage**.

★ Basic Explanation

From syllabus:

Feature	MAC	Digital Signature
Key type	Symmetric	Asymmetric
Verification	Shared secret	Public key
Speed	Fast	Slower
Non-repudiation	No	Yes

★ Exam Definition

A MAC uses a shared secret key to provide authentication and integrity, whereas a digital signature uses a private-public key pair and also provides non-repudiation.

★ Real-Life Analogy

MAC → Secret handshake

Digital Signature → Notarized signature

MA

-

-

MAC

Ser

Dig

Ser

-

MA

MA

Dig

$$S =$$

◆ STEP-4: NUMERICAL EXAMPLE

MAC example:

$K = 10$

$M = 5$

$MAC = K + M = 15$

Digital signature example:

Hash = 20

Private key encrypt $\rightarrow$ signature.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

MAC uses a shared secret key and provides authentication and integrity but no non-repudiation. Digital signatures use asymmetric keys and provide authentication, integrity, and non-repudiation.

◆ STEP-6: EXAM TIP

- Draw comparison table
 - Write symmetric vs asymmetric
 - Mention non-repudiation
 - High-frequency PYQ
-

◆ Q12. (Chapter-8 Key Management)

Explain Diffie-Hellman key exchange with example and diagram.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Two strangers create a shared secret without meeting.

Magic of math.

★ Basic Explanation

Diffie-Hellman allows secure key exchange over public channel.

Public values:

- Prime p
- Generator g

Each user picks private number.

★ Exam Definition

Diffie-Hellman is a key agreement protocol that enables two parties to generate a shared secret over an insecure channel using modular exponentiation.

★ Real-Life Analogy

Mixing colors secretly:

- Public base color
- Private mix $\rightarrow$ same final color

★ Practical Use

- HTTPS
- VPN
- Secure messaging

★ Exam Points

- Uses modular arithmetic
- Prevents eavesdropping

◆ STEP-2: DIAGRAM

Public: p, g

Alice:	Bob:
Choose private a	Choose private b
Compute $A = g^a \bmod p$	
----->	Compute $B = g^b \bmod p$
	<-----

Shared key:

Alice: $K = B^a \bmod p$

Bob: $K = A^b \bmod p$

Same key!

🔍 Label Explanation

- a, b = private keys
 - A, B = public values
 - K = shared secret
-

◆ STEP-3: FORMULAS

$$A = g^a \bmod p$$

$$B = g^b \bmod p$$

Shared key:

$$K = (g^b)^a \bmod p = g^{ab} \bmod p$$

◆ STEP-4: FULL NUMERICAL EXAMPLE

Given:

$$p = 23$$

$$g = 5$$

Alice private $a = 6$

Bob private $b = 15$

Step 1: Alice public

$$A = 5^6 \bmod 23$$

$$5^6 = 15625$$

$$15625 \bmod 23 = 8$$

Step 2: Bob public

$$B = 5^{15} \bmod 23$$

$$\text{Result} = 19$$

Step 3: Shared key

Alice:

$$K = 19^6 \bmod 23 = 2$$

Bob:

$$K = 8^{15} \bmod 23 = 2$$

Same key = 2 

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Diffie-Hellman enables two parties to generate a shared secret key using modular exponentiation

without transmitting the key itself.

◆ STEP-6: EXAM TIP

- Always show mod calculations
 - Write $A = g^a \text{ mod } p$ formula
 - Draw exchange arrows
 - Common numerical question
-

◆ Q13. (Chapter-8 Key Management)

Explain Kerberos authentication with diagram.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Central trusted authority verifies everyone.

Like college ID system.

★ Basic Explanation

Kerberos uses:

- Authentication Server (AS)
- Ticket Granting Server (TGS)

Users get tickets to access services.

★ Exam Definition

Kerberos is a network authentication protocol that uses symmetric key cryptography and a trusted third party to authenticate users via tickets.

★ Real-Life Analogy

Movie ticket system:

- Buy ticket
 - Show ticket for entry
-

★ Practical Use

- Windows Active Directory
 - Enterprise login
-

★ Exam Points

- Ticket-based
 - Time-stamped tickets
 - Prevents replay attacks
-

◆ STEP-2: DIAGRAM

```
User → AS (Authentication Server)
      |
      | Ticket for TGS
      v
User → TGS (Ticket Granting Server)
      |
      | Ticket for Service
      v
User → Service Server
      |
      Access granted
```

🔍 Label Explanation

- AS = initial authentication
 - TGS = ticket generator
 - Ticket = encrypted credential
-

◆ STEP-3: FORMULAS

$\text{Ticket} = \text{Encrypt}(K_{\text{server}}, \{\text{User}, \text{Timestamp}\})$

Where:

K_{server} = server secret key

◆ STEP-4: NUMERICAL

Conceptual protocol — no math.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Kerberos is a trusted third-party authentication protocol that uses tickets issued by AS and TGS to securely authenticate users and services.

◆ STEP-6: EXAM TIP

- Draw 3-server diagram
 - Write AS, TGS roles
 - Mention tickets
 - PYQ favorite
-

◆ Q14. (Chapter-7 Entity Authentication)

Explain password-based authentication methods.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Simplest authentication = password.

★ Basic Explanation

Types:

1. Plaintext password
 2. Hashed password
 3. Salted password
 4. Hybrid authentication
-

★ Exam Definition

Password-based authentication verifies identity using secret passwords stored in plaintext, hashed, or salted form.

★ Real-Life Analogy

Phone lock PIN.

★ Practical Use

- Login systems

- ATM PIN
-

★ Exam Points

- Hash improves security
 - Salt prevents dictionary attacks
-

◆ STEP-2: DIAGRAM

User enters password

|

v

Hash(password + salt)

|

Compare with stored hash

|

Match → Access

🔍 Label Explanation

- Salt = random value
 - Hash = secure storage
-

◆ STEP-3: FORMULAS

Stored password:

$H(P \parallel \text{Salt})$

Where:

P = password

◆ STEP-4: NUMERICAL EXAMPLE

Password = admin

Salt = 123

Stored:

$H(\text{admin}123)$

Attacker cannot guess easily.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Password-based authentication verifies identity using stored passwords in plaintext, hashed, or salted forms, with salting improving resistance against dictionary attacks.

◆ STEP-6: EXAM TIP

- Mention salting
 - Write 3 methods
 - Draw hash flow
 - Very common question
-

◆ Q15. (Chapter-9 DoS Attack)

Classify different types of DoS attacks.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Different ways to overload a system.

★ Basic Explanation

From syllabus:

Types:

1. Flooding attacks
 2. Protocol attacks
 3. Application attacks
-

★ Exam Definition

DoS attacks are classified based on attack layer and method such as flooding, protocol exploitation, and application-level attacks.

★ Real-Life Analogy

Traffic jam types:

- Too many cars
- Broken signals

- Blocked roads

★ Practical Use

- Website crashes
- Network disruption

★ Exam Points

- SYN flood
- UDP flood
- ICMP flood

◆ STEP-2: DIAGRAM

DoS Attacks

```
|  
| --- Flooding (SYN, UDP, ICMP)  
|  
| --- Protocol Exploit  
|  
| --- Application Attack
```

🔍 Label Explanation

- Flooding = bandwidth overload
- Protocol = handshake exploit
- Application = server crash

◆ STEP-3: FORMULAS

No formulas.

◆ STEP-4: NUMERICAL EXAMPLE

Server bandwidth = 1 Gbps

Attack traffic = 5 Gbps

Result → Server unavailable.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

DoS attacks are classified into flooding attacks, protocol exploitation attacks, and application-layer

attacks, each targeting system availability using different techniques.

◆ **STEP-6: EXAM TIP**

- Draw classification tree
- Write 3 examples
- Mention availability attack
- Easy scoring question

◆ Q16. (Chapter-4 Digital Signature)

Explain attacks on digital signatures.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Hackers try to:

- Forge signatures
- Modify signed data

This breaks trust.

★ Basic Explanation

Common attacks:

1. Key compromise
2. Signature forgery

3. Replay attack
4. Hash collision attack

★ Exam Definition

Attacks on digital signatures aim to forge or misuse signatures by exploiting key compromise, collisions, or replay mechanisms.

★ Real-Life Analogy

Forging someone's handwritten signature.

★ Practical Use

- Fake software updates
- Fraud transactions

★ Important Exam Points

- Private key theft is worst attack
- Collision attacks target hash

◆ STEP-2: DIAGRAM

Normal:

Sender → Sign → Message + Signature → Receiver

Attack:

Attacker steals private key

|

Creates fake signature

|

Receiver thinks valid

🔍 Label Explanation

- Private key theft = root problem
- Fake signature = forgery

◆ STEP-3: FORMULAS

Signature:

$S = \text{Encrypt_private}(H(M))$

Forgery:

Attacker computes fake S

◆ **STEP-4: NUMERICAL EXAMPLE**

If attacker finds:

$$H(M1) = H(M2)$$

Then:

Signature of M1 works for M2 → attack.

◆ **STEP-5: FINAL ANSWER**

👉 **FINAL ANSWER:**

Digital signature attacks include key compromise, signature forgery, replay attacks, and hash collisions that undermine authenticity and integrity.

◆ **STEP-6: EXAM TIP**

- Write 4 attack types
 - Mention collision clearly
 - Use word “forgery”
 - Short but scoring question
-

◆ **Q17. (Chapter-5 Cryptographic Hash Functions)**

Explain preimage, second preimage, and collision resistance.

◆ **STEP-1: THEORY EXPLANATION**

★ **Intuition**

Hash should be impossible to reverse or duplicate.

★ **Basic Explanation**

Three core properties:

1 **Preimage resistance**

Given hash → cannot find input.

2 Second preimage resistance

Given $M_1 \rightarrow$ cannot find M_2 with same hash.

3 Collision resistance

Cannot find any two messages with same hash.

★ Exam Definition

These are security properties that ensure cryptographic strength of hash functions against inversion and collision attacks.

★ Real-Life Analogy

Fingerprint uniqueness.

★ Practical Use

- Blockchain security
- Password hashing

★ Exam Points

Collision resistance strongest property.

◆ STEP-2: DIAGRAM

Preimage:

$H(M) \rightarrow$ Given hash $\rightarrow$ find M (hard)

Second Preimage:

$M_1 \rightarrow H(M_1)$

Find $M_2 \neq M_1$ but $H(M_2)=H(M_1)$

Collision:

Find M_1, M_2 where $H(M_1)=H(M_2)$

🔍 Label Explanation

- M_1, M_2 = different messages
- $H(M)$ = hash digest

◆ STEP-3: FORMULAS

$H(M) \rightarrow$ digest

Collision:

Find $M1 \neq M2$ such that:

$$H(M1) = H(M2)$$

◆ STEP-4: NUMERICAL EXAMPLE

SHA-256 $\rightarrow$ 256-bit output

$$\text{Total hashes} = 2^{256}$$

Collision complexity $\approx 2^{128}$ (birthday paradox).

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Preimage resistance prevents finding input from hash, second preimage resistance prevents finding another input with same hash, and collision resistance prevents finding any two inputs with identical hashes.

◆ STEP-6: EXAM TIP

- Write definitions clearly
- Draw comparison diagram
- Mention birthday attack
- Very common theory question

◆ Q18. (Chapter-6 Message Integrity & Authentication)

Explain Random Oracle Model.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Assume hash behaves like a **perfect random function**.

★ Basic Explanation

Random Oracle Model treats hash as:

- Black box
- Random outputs

Used in theoretical proofs.

★ Exam Definition

The Random Oracle Model is an abstraction where hash functions are modeled as ideal random functions producing unpredictable outputs.

★ Real-Life Analogy

Lottery machine — unpredictable numbers.

★ Practical Use

- Security proofs
 - Digital signatures
-

★ Exam Points

- Used in theoretical cryptography
 - Not real but approximation
-

◆ STEP-2: DIAGRAM

Input M

|

v

[Random Oracle]

|

Random output $H(M)$

Same input $\rightarrow$ same output

Different input $\rightarrow$ unpredictable output

🔍 Label Explanation

- Oracle = ideal hash function
 - Deterministic but random-looking
-

◆ STEP-3: FORMULAS

$H(M)$ = Random output

Properties:

- Deterministic
 - Uniform distribution
-

◆ STEP-4: NUMERICAL EXAMPLE

Input:

HELLO → 9A2F

HELLo → B71C

Completely different outputs.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Random Oracle Model treats hash functions as ideal random functions producing unpredictable outputs, mainly used in cryptographic security proofs.

◆ STEP-6: EXAM TIP

- Write “ideal hash assumption”
 - Draw black-box diagram
 - Mention theoretical model
 - Short but conceptual question
-

◆ Q19. (Chapter-8 Key Management)

Explain Key Distribution Center (KDC) with diagram.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Instead of everyone sharing keys with everyone, use a trusted middleman.

★ Basic Explanation

KDC reduces key explosion problem.

Without KDC:

N users → N^2 keys.

With KDC:

Each user shares key only with KDC.

★ Exam Definition

A Key Distribution Center is a trusted third party that securely distributes session keys to communicating entities.

★ Real-Life Analogy

Bank locker manager distributing keys.

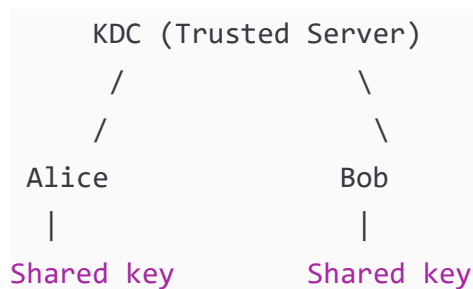
★ Practical Use

- Kerberos
 - Enterprise security
-

★ Exam Points

- Solves key management problem
 - Creates session keys
-

◆ STEP-2: DIAGRAM



Alice asks KDC → session key
KDC sends secure session key

🔍 Label Explanation

- KDC = trusted authority
 - Session key = temporary key
-

◆ STEP-3: FORMULAS

Session key:

K_{AB} = generated by KDC

Encrypted:

$E(K_A, K_{AB})$

Where:

KA = Alice-KDC key

◆ **STEP-4: NUMERICAL EXAMPLE**

3 users:

Without KDC $\rightarrow 3^2 = 9$ keys

With KDC $\rightarrow$ only 3 keys.

Huge reduction.

◆ **STEP-5: FINAL ANSWER**

👉 **FINAL ANSWER:**

KDC is a trusted third party that generates and distributes session keys, reducing the total number of required keys in a network.

◆ **STEP-6: EXAM TIP**

- Write N^2 problem
 - Draw triangle diagram
 - Mention session keys
 - Frequently asked
-

◆ **Q20. (Chapter-9 DoS Attack)**

Explain TCP three-way handshake and its role in DoS attacks.

◆ **STEP-1: THEORY EXPLANATION**

★ **Intuition**

TCP needs 3 steps to connect.

Attackers exploit this.

★ **Basic Explanation**

Normal handshake:

1. SYN
2. SYN-ACK

3. ACK

★ Exam Definition

The TCP three-way handshake establishes reliable connection between client and server and is exploited in SYN flood attacks.

★ Real-Life Analogy

Phone call confirmation:
Call → ring → answer.

★ Practical Use

Reliable communication.

★ Exam Points

- Connection-oriented protocol
 - Used by HTTP, FTP
-

◆ STEP-2: DIAGRAM

Client → SYN → Server
Server → SYN-ACK → Client
Client → ACK → Connection Established

🔍 Label Explanation

- SYN = synchronize
 - ACK = acknowledgment
-

◆ DoS Exploit Diagram

Attacker → SYN → Server
Server → SYN-ACK → (No reply)
Many half-open connections
Server overload

◆ STEP-3: FORMULAS

No math.

◆ STEP-4: NUMERICAL EXAMPLE

Server backlog queue = 50

Attacker sends 200 SYN requests.

Result:

Queue full → real users blocked.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

TCP three-way handshake establishes connections using SYN, SYN-ACK, and ACK steps, but attackers exploit it in SYN flood attacks by creating half-open connections.

◆ STEP-6: EXAM TIP

- Draw handshake diagram
- Write 3 steps clearly
- Mention SYN flood
- High-probability question

◆ Q21. (Chapter-8 Key Management)

Explain Needham-Schroeder protocol with diagram.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Two users want a secret key.

They use a trusted server.

This is Needham-Schroeder.

★ Basic Explanation

It is a symmetric key authentication protocol using a KDC.

Uses:

- Nonces (random numbers)
 - Session keys
-

★ Exam Definition

Needham-Schroeder protocol is a symmetric key authentication protocol that uses a trusted server and nonce values to securely distribute session keys.

★ Real-Life Analogy

Mediator introducing two people safely.

★ Practical Use

- Basis of Kerberos
 - Secure authentication systems
-

★ Exam Points

- Uses nonce to prevent replay
 - Session key distribution
-

◆ STEP-2: DIAGRAM

1. Alice → KDC : ID_A, ID_B, RA
 2. KDC → Alice :
{RA, ID_B, K_AB, Ticket_B} encrypted with KA
 3. Alice → Bob :
Ticket_B
 4. Bob → Alice : RB
 5. Alice → Bob : f(K_AB, RB)
-

🔍 Label Explanation

- RA, RB = nonces
 - K_AB = session key
 - Ticket_B = encrypted ticket
-

◆ STEP-3: FORMULAS

Ticket:

$E(KB, \{K_{AB}, ID_A\})$

Where:

KB = Bob's key

K_AB = session key

◆ STEP-4: NUMERICAL EXAMPLE

RA = 101

RB = 202

Alice proves identity by encrypting RB using session key.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Needham-Schroeder protocol uses a trusted server and nonce values to securely distribute a session key between two parties and prevent replay attacks.

◆ STEP-6: EXAM TIP

- Write 5-step flow
 - Draw nonce exchange
 - Mention replay protection
 - Often asked with Kerberos
-

◆ Q22. (Chapter-7 Entity Authentication)

Compare different entity authentication methods.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

How systems verify identity:

- Something you know
 - Something you have
 - Something you are
-

★ Basic Explanation

1 Knowledge-based

Password, PIN

2 Possession-based

Smart cards

3 Biometric

Fingerprint, face

★ Exam Definition

Entity authentication methods verify identity using knowledge, possession, or inheritance-based factors.

★ Real-Life Analogy

ATM:

Card + PIN.

★ Practical Use

- Multi-factor authentication
 - Secure login
-

★ Exam Points

Multi-factor = strongest.

◆ STEP-2: DIAGRAM

Authentication Methods

```
|  
|--- Knowledge (Password)  
|  
|--- Possession (Smart card)  
|  
|--- Inherence (Biometric)
```

🔍 Label Explanation

- Knowledge = secret known
 - Possession = physical object
 - Inherence = body feature
-

◆ **STEP-3: FORMULAS**

Not mathematical.

◆ **STEP-4: NUMERICAL EXAMPLE**

Bank login:

Card + PIN = 2-factor authentication.

◆ **STEP-5: FINAL ANSWER**

👉 **FINAL ANSWER:**

Entity authentication methods include knowledge-based, possession-based, and biometric methods, with multi-factor authentication providing higher security.

◆ **STEP-6: EXAM TIP**

- Write 3 categories clearly
 - Give examples
 - Mention MFA
 - Easy scoring question
-

◆ **Q23. (Chapter-5 Cryptographic Hash Functions)**

Explain padding in SHA-512 with example.

◆ **STEP-1: THEORY EXPLANATION**

★ **Intuition**

Hash works on fixed-size blocks.

So message must be padded.

★ **Basic Explanation**

SHA-512 requires message length multiple of 1024 bits.

Steps:

1. Add '1' bit
2. Add zeros
3. Add length field

★ Exam Definition

Padding in SHA-512 ensures the message length becomes a multiple of 1024 bits by adding a 1-bit, zeros, and a 128-bit length field.

★ Real-Life Analogy

Adding spaces to fill a form.

★ Practical Use

Ensures proper block processing.

★ Exam Points

- 128-bit length field
- Merkle-Damgård usage

◆ STEP-2: DIAGRAM

Original Message

|

Add 1 bit

|

Add 0 padding

|

Add 128-bit length

|

1024-bit block ready

🔍 Label Explanation

- Padding ensures fixed block size
- Length field prevents extension attacks

◆ STEP-3: FORMULAS

Total length:

$$|M| + 1 + k + 128 \equiv 0 \pmod{1024}$$

Where:

k = number of zeros

◆ STEP-4: NUMERICAL EXAMPLE

Message length = 800 bits

We need:

$800 + 1 + k + 128 = \text{multiple of } 1024$

Solve:

$929 + k = 1024$

$k = 95$ zeros

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

SHA-512 padding adds a 1-bit, zero padding, and a 128-bit length field so that the final message length becomes a multiple of 1024 bits.

◆ STEP-6: EXAM TIP

- Write padding formula
 - Show one calculation
 - Mention 128-bit length field
 - Very scoring numerical
-

◆ Q24. (Chapter-4 Digital Signature)

Explain digital signature verification process.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Receiver checks:

Did sender really sign this?

★ Basic Explanation

Verification steps:

1. Hash received message
 2. Decrypt signature with public key
 3. Compare hashes
-

★ Exam Definition

Digital signature verification involves validating the signature using the sender's public key and comparing computed and decrypted hashes.

★ Real-Life Analogy

Comparing original signature with sample.

★ Practical Use

- Email verification
 - Blockchain validation
-

★ Exam Points

Public key used in verification.

◆ STEP-2: DIAGRAM

Receiver gets:

Message M + Signature S

Step 1:

Compute $H(M)$

Step 2:

Decrypt S using public key $\rightarrow H1$

Step 3:

Compare $H(M)$ and $H1$

Match $\rightarrow$ Valid

🔍 Label Explanation

- $H(M)$ = new hash
 - $H1$ = extracted hash
-

◆ STEP-3: FORMULAS

Verification:

$H(M) = \text{Decrypt_public}(S)$

◆ **STEP-4: NUMERICAL EXAMPLE**

$H(M) = 45$

Decrypt signature $\rightarrow 45$

Match $\rightarrow$ valid signature.

◆ **STEP-5: FINAL ANSWER**

👉 **FINAL ANSWER:**

Digital signature verification involves hashing the received message and comparing it with the decrypted signature hash using the sender's public key.

◆ **STEP-6: EXAM TIP**

- Draw verification flow
 - Write 3 steps clearly
 - Mention public key
 - Often asked with signing process
-

◆ **Q25. (Chapter-9 DoS Attack)**

Explain methods to prevent DoS attacks.

◆ **STEP-1: THEORY EXPLANATION**

★ **Intuition**

Stop attackers from flooding resources.

★ **Basic Explanation**

Common prevention methods:

- 1 Firewalls
 - 2 Rate limiting
 - 3 SYN cookies
 - 4 Load balancing
-

★ Exam Definition

DoS prevention involves techniques that detect and mitigate excessive traffic or malformed requests to maintain service availability.

★ Real-Life Analogy

Traffic police controlling congestion.

★ Practical Use

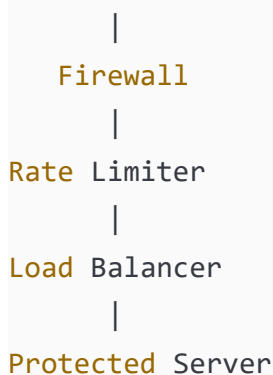
- Cloud security
 - Website protection
-

★ Exam Points

- Prevention > detection
 - Use distributed defense
-

◆ STEP-2: DIAGRAM

Internet Traffic



🔍 Label Explanation

- Firewall = filter traffic
 - Rate limiter = control speed
 - Load balancer = distribute load
-

◆ STEP-3: FORMULAS

No math.

◆ STEP-4: NUMERICAL EXAMPLE

Without load balancing:

1 server handles 10k requests → crash

With 5 servers:

Each handles 2k → stable.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

DoS attacks can be prevented using firewalls, rate limiting, SYN cookies, and load balancing to filter malicious traffic and maintain availability.

◆ STEP-6: EXAM TIP

- Write 4 methods
- Draw protection pipeline
- Mention availability
- Short but scoring answer

◆ Q26. (Chapter-4 Digital Signature)

Explain existential forgery and selective forgery in digital signatures.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Forgery = fake signature creation.

Two types:

- Random fake
 - Targeted fake
-

★ Basic Explanation

From syllabus:

1 Existential forgery

Attacker creates any valid message-signature pair.

2 Selective forgery

Attacker creates signature for a chosen message.

★ Exam Definition

Existential forgery allows generating any valid signed message, while selective forgery allows forging a signature for a specific target message.

★ Real-Life Analogy

Existential → Random fake cheque

Selective → Fake signature on a specific contract

★ Practical Use

Evaluates signature scheme strength.

★ Exam Points

Selective forgery = more dangerous.

◆ STEP-2: DIAGRAM

Existential Forgery:

Attacker → Creates random (M, S) pair

Selective Forgery:

Attacker chooses message M

|

Creates fake signature S

🔍 Label Explanation

- M = message

- S = signature

◆ STEP-3: FORMULAS

Forgery condition:

$\text{Verify}(M, S) = \text{TRUE}$ (even if fake)

◆ STEP-4: NUMERICAL EXAMPLE

If attacker finds:

$H(M1) = H(M2)$

Signature of M1 works for M2 → selective forgery.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Existential forgery creates any valid signed message, while selective forgery targets forging a signature for a specific chosen message.

◆ STEP-6: EXAM TIP

- Write both definitions clearly
- Mention which is stronger attack
- Short scoring question

◆ Q27. (Chapter-5 Cryptographic Hash Functions)

Differentiate between hashing and encryption.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Encryption = reversible

Hashing = irreversible

★ Basic Explanation

Feature	Hashing	Encryption
Reversible	No	Yes

Feature	Hashing	Encryption
Key needed	No	Yes
Output size	Fixed	Variable

★ Exam Definition

Hashing is a one-way transformation producing fixed-length digests, while encryption is a reversible process using keys for confidentiality.

★ Real-Life Analogy

Hash → fingerprint

Encryption → locked box

★ Practical Use

Hash:

Passwords

Encryption:

Secure communication

★ Exam Points

Hash = integrity

Encryption = confidentiality

◆ STEP-2: DIAGRAM

Hash:

Message → Hash → Digest (cannot reverse)

Encryption:

Plaintext → Encrypt(K) → Ciphertext

Ciphertext → Decrypt(K) → Plaintext

🔍 Label Explanation

- K = key
- Digest = hash output

◆ STEP-3: FORMULAS

Hash:
 $H(M)$

Encryption:
 $C = E(K, P)$

Decryption:
 $P = D(K, C)$

◆ STEP-4: NUMERICAL EXAMPLE

Password:
 $\text{Hash}(\text{admin}) \rightarrow \text{stored}$

Encryption:
 $\text{admin} \rightarrow \text{encrypted} \rightarrow \text{decrypted back.}$

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Hashing is a one-way function used for integrity, while encryption is a reversible key-based process used for confidentiality.

◆ STEP-6: EXAM TIP

- Draw comparison table
 - Mention reversible vs irreversible
 - Very common PYQ
-

◆ Q28. (Chapter-4 Digital Signature)

State and explain properties of digital signatures.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

A good signature must be:

- Unique
 - Hard to forge
 - Easy to verify
-

★ Basic Explanation

Properties:

- 1 Depends on message
 - 2 Uses sender's private key
 - 3 Easy to verify
 - 4 Hard to forge
-

★ Exam Definition

Digital signatures must ensure authenticity, integrity, non-repudiation, and computational infeasibility of forgery.

★ Real-Life Analogy

Handwritten signature uniqueness.

★ Practical Use

Legal documents, blockchain.

★ Exam Points

Non-repudiation is key property.

◆ STEP-2: DIAGRAM

Message M → Hash → Sign with Private Key → Signature

Receiver:

Verify using Public Key

🔍 Label Explanation

- Private key = uniqueness
 - Public key = verification
-

◆ STEP-3: FORMULAS

$S = \text{Encrypt_private}(H(M))$

◆ STEP-4: NUMERICAL EXAMPLE

If two messages differ:

$M1 \neq M2 \rightarrow H(M1) \neq H(M2)$

Signature changes.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Digital signatures must be message-dependent, unique to sender, easy to verify, hard to forge, and provide authenticity and non-repudiation.

◆ STEP-6: EXAM TIP

- Write 4 properties
 - Mention non-repudiation
 - Easy scoring
-

◆ Q29. (Chapter-8 Key Management)

Explain security issues in Diffie-Hellman key exchange.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Diffie-Hellman is secure but not perfect.

Main problem = Man-in-the-Middle.

★ Basic Explanation

Issues:

- 1 Man-in-the-middle attack
 - 2 No authentication
 - 3 Public key tampering
-

★ Exam Definition

Diffie-Hellman is vulnerable to man-in-the-middle attacks due to lack of authentication of exchanged public values.

★ Real-Life Analogy

Someone intercepting letters and replacing them.

★ Practical Use

Solved using certificates.

★ Exam Points

Authentication required.

◆ STEP-2: DIAGRAM

Alice → A → Attacker → Bob
Attacker sends fake values

Alice and Bob think secure
But attacker knows key

🔍 Label Explanation

- Attacker acts as relay
 - Creates two separate keys
-

◆ STEP-3: FORMULAS

Fake exchange:

Alice shares key with attacker

Bob shares key with attacker

◆ STEP-4: NUMERICAL EXAMPLE

Alice key with attacker = 10

Bob key with attacker = 20

Attacker decrypts everything.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Diffie-Hellman is vulnerable to man-in-the-middle attacks due to lack of authentication, which can be mitigated using certificates or digital signatures.

◆ STEP-6: EXAM TIP

- Write MITM clearly
 - Draw attacker in middle
 - Mention solution (certificates)
 - High exam probability
-

◆ Q30. (Chapter-1 Overall Revision Concept)

Explain relationship between security attacks, services, and mechanisms.

◆ STEP-1: THEORY EXPLANATION

★ Intuition

Security is like war:

- Attackers attack
 - Services defend
 - Mechanisms are weapons
-

★ Basic Explanation

From network security intro:

1 Security Attack

Threat to system

2 Security Service

Protection goal

3 Security Mechanism

Implementation tool

★ Exam Definition

Security attacks compromise systems, security services provide protection objectives, and security mechanisms implement those protections.

★ Real-Life Analogy

Thief → attack

Police → service

Weapons → mechanism

★ Practical Use

OSI security architecture.

★ Exam Points

All three interconnected.

◆ STEP-2: DIAGRAM

Security Attack

↓

Security **Service** (Confidentiality, Integrity)

↓

Security **Mechanism** (Encryption, Hashing, Signatures)

🔍 Label Explanation

- Attack triggers need for service
 - Mechanism implements service
-

◆ STEP-3: FORMULAS

Conceptual.

◆ STEP-4: NUMERICAL EXAMPLE

Attack: Data theft

Service: Confidentiality

Mechanism: Encryption.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Security attacks threaten systems, security services define protection goals, and security mechanisms implement those protections using cryptographic techniques.

◆ STEP-6: EXAM TIP

- Draw 3-level diagram
 - Give one example chain
 - Easy conceptual question
-
-

